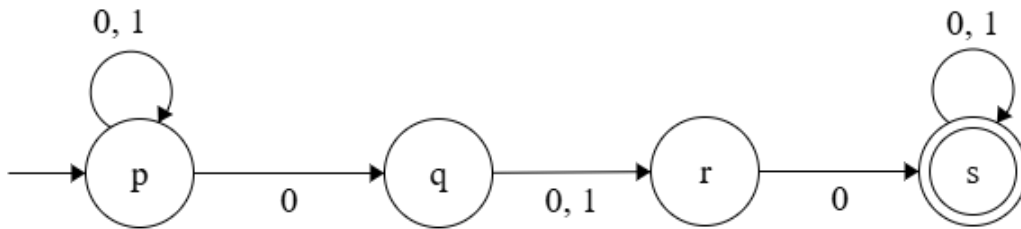


Oefeningen talen en automaten - Reeks 3

Oefeningen bij 2.3.5 en 2.4

Reeks 3

1.



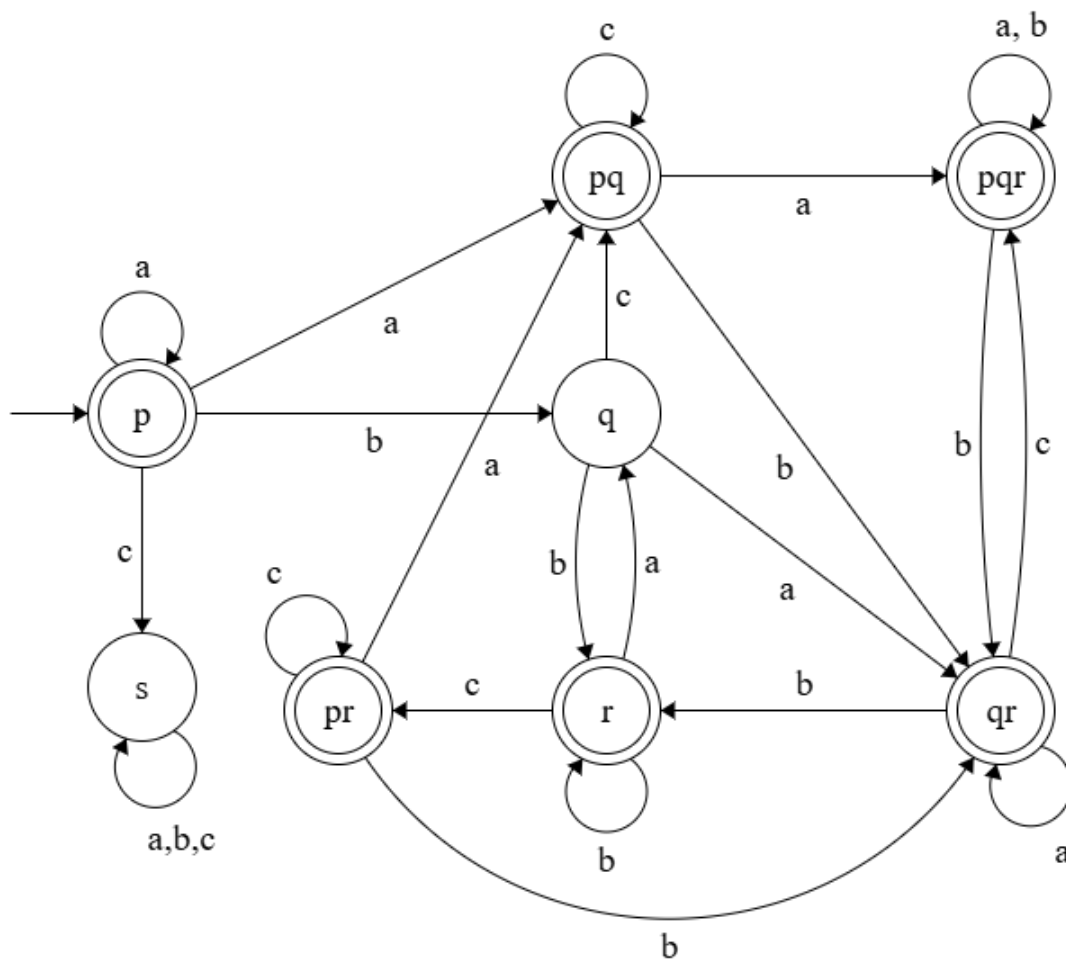
2.

DFA

Transitie Tabel

	a	b	c
→ *p	pq	q	s
*pq	pqr	qr	pq
q	qr	r	pq
s	s	s	s
*pqr	pqr	qr	pqr
*qr	qr	r	pqr
*r	q	r	pr
*pr	pq	qr	pr

Transitiediagram



3.

transitietabel die bij deze NFA hoort

		0	1
→	*q0	∅	{q1, q}
q1		{q}	∅
*q		∅	∅

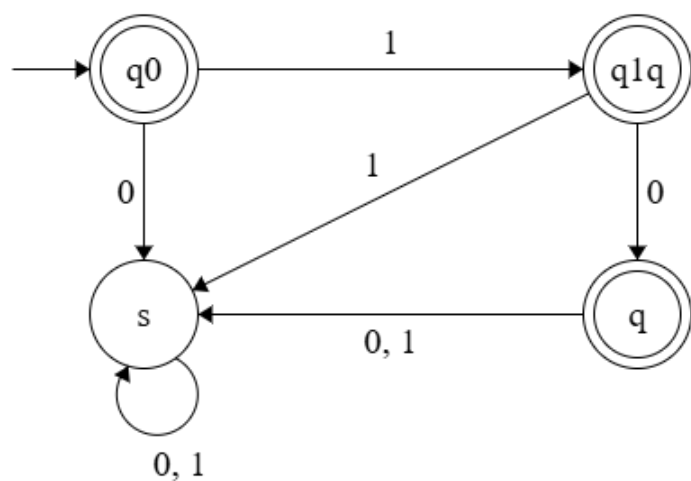
DFA

$D = \{\{q0, q1q, q, s\}, \{0, 1\}, \{q0, q1q, q, s\} \times \{0, 1\} \rightarrow \{q0, q1q, q, s\}, \{q0\}, \{q0, q1q, q\}\}$

Transitietabel

	0	1
→ *q0	s	q1q
*q1q	q	s
*q	s	s
s	s	s

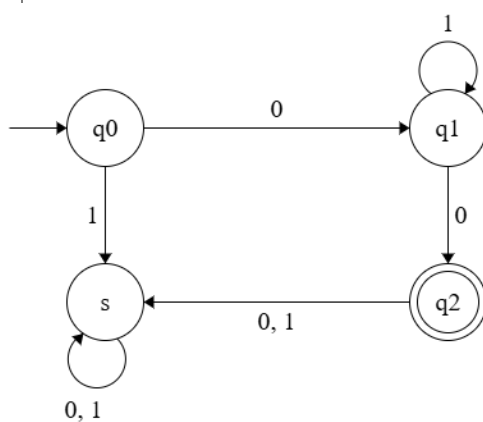
Transitiedigram



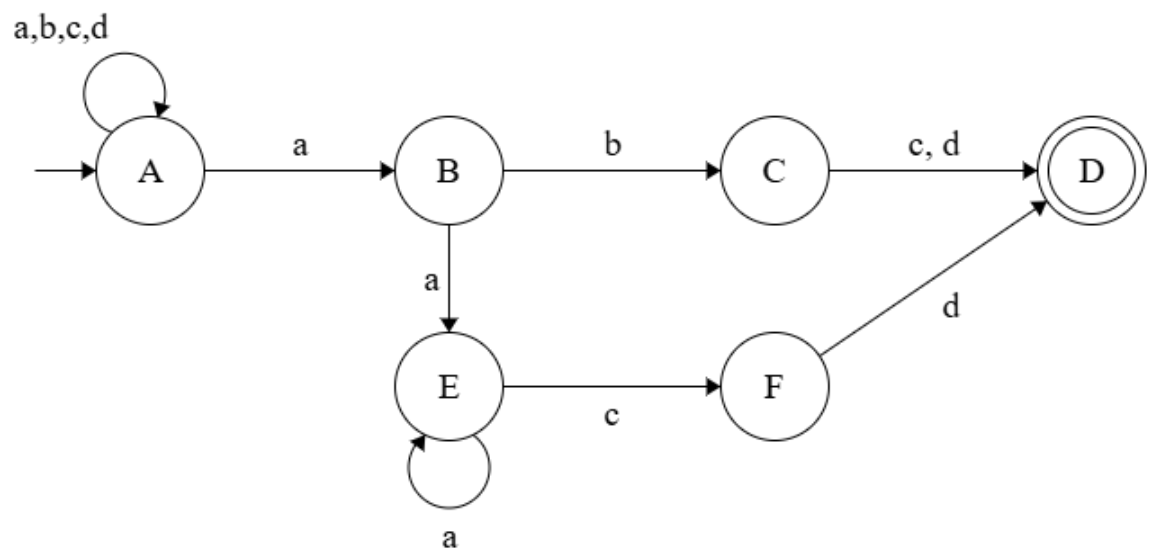
4.
Het diagram stelt een NFA, want niet alle symbolen worden gebruikt bij ieder toestand.

DFA

	0	1
→ q0	q1	s
q1	q2	q1
*q2	s	s
s	s	s



5.

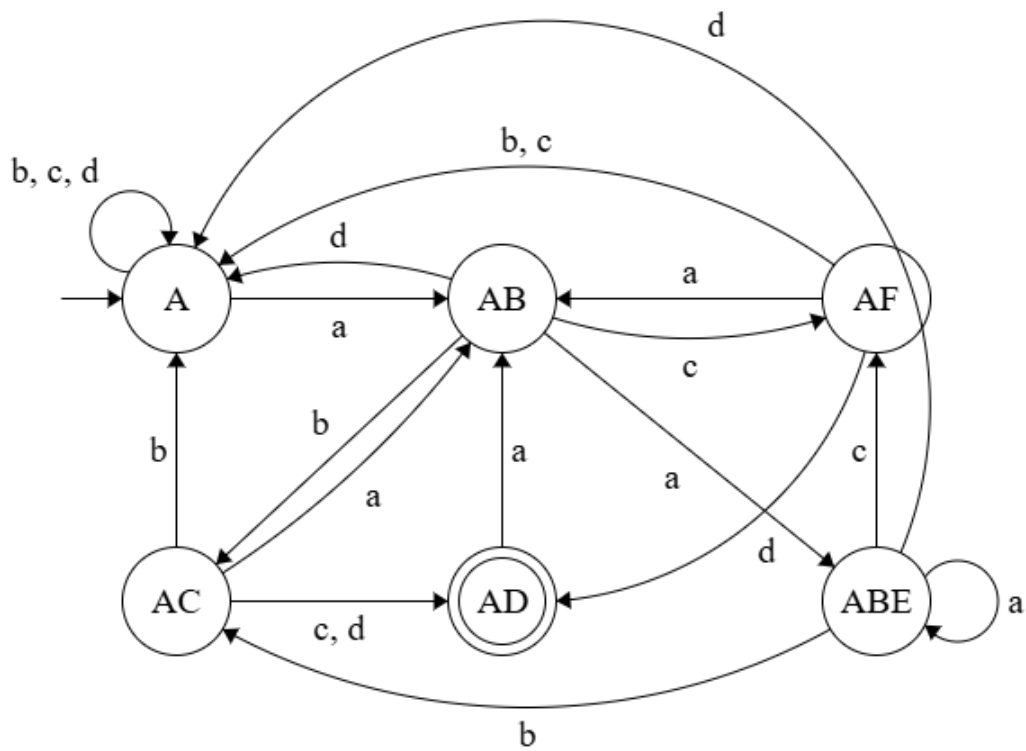


	a	b	c	d
→ A	{A, B}	{A}	{A}	{A}
B	{E}	{C}	∅	∅
C	∅	∅	{D}	{D}
*D	∅	∅	∅	∅
E	{E}	∅	{F}	∅
F	∅	∅	∅	{D}

6.

DFA

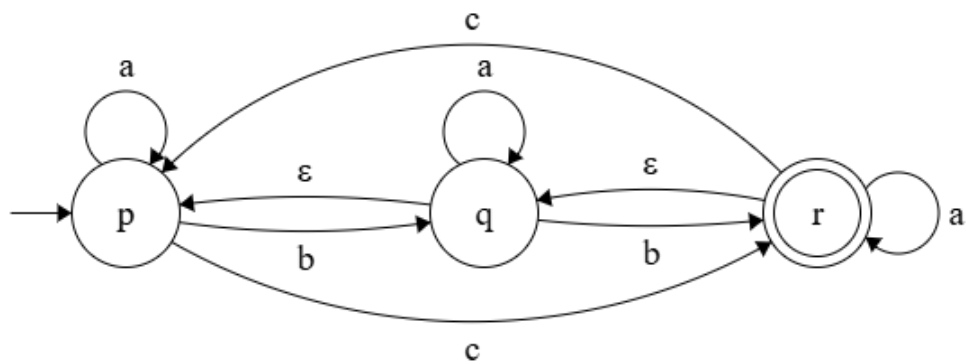
	a	b	c	d
→ A	AB	A	A	A
AB	ABE	AC	A	A
AC	AB	A	AD	AD
*AD	AB	A	A	A
AF	AB	A	A	AD
ABE	ABE	AC	AF	A



Reeks 4

1.

a.



b.

$$Eclosure(p) = \{p\}$$

$$Eclosure(q) = \{p, q\}$$

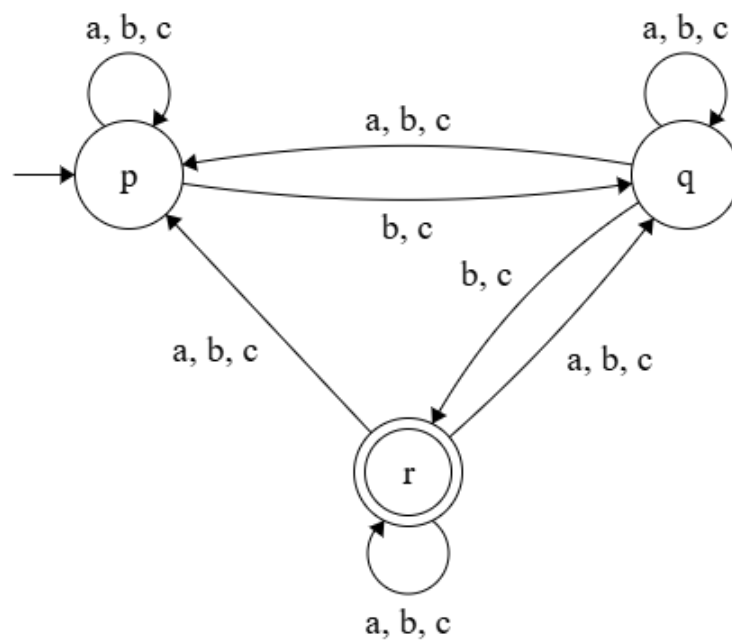
$$\text{Eclosure}(r) = \{p, q, r\}$$

c.

	ϵ^*	a	ϵ^*		ϵ^*	b	ϵ^*
p	$\{p\}$	$\{p\}$	$\{p\}$	p	$\{p\}$	$\{q\}$	$\{p, q\}$
q	$\{p, q\}$	$\{p, q\}$	$\{p, q\}$	q	$\{p, q\}$	$\{q, r\}$	$\{p, q, r\}$
r	$\{p, q, r\}$	$\{p, q, r\}$	$\{p, q, r\}$	r	$\{p, q, r\}$	$\{q, r\}$	$\{p, q, r\}$
	ϵ^*	c			ϵ^*		
p	$\{p\}$	$\{r\}$			$\{p, q, r\}$		
q	$\{p, q\}$	$\{r\}$			$\{p, q, r\}$		
r	$\{p, q, r\}$	$\{p, r\}$			$\{p, q, r\}$		

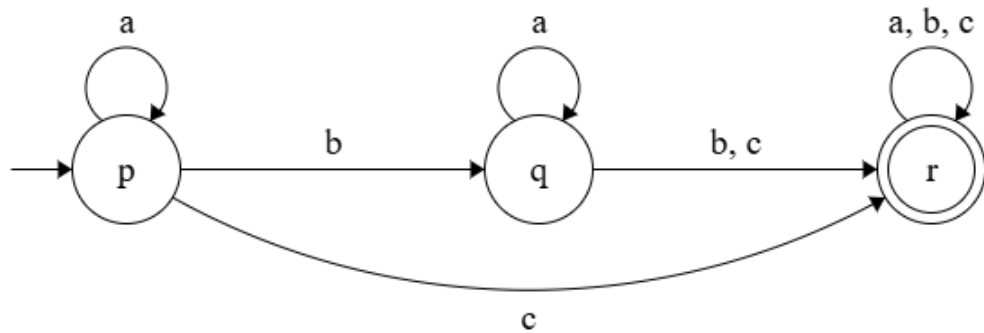
NFA

	a	b	c
→ p	$\{p\}$	$\{p, q\}$	$\{p, q, r\}$
q	$\{p, q\}$	$\{p, q, r\}$	$\{p, q, r\}$
*r	$\{p, q, r\}$	$\{p, q, r\}$	$\{p, q, r\}$



DFA

	a	b	c
→ p	p	q	r
q	q	r	r
*r	r	r	r



2.

	ϵ	a	b	c
→ p	$\emptyset$	{p, q}	$\emptyset$	$\emptyset$
*q	{r}	$\emptyset$	{p, q}	$\emptyset$
s	{r}	$\emptyset$	$\emptyset$	{s}
r	$\emptyset$	{r, s}	$\emptyset$	{q, r}

E-closure

	ϵ^*	a	ϵ^*		ϵ^*	b	ϵ^*
p	{p}	{p, q}	{p, q, r}	p	{p}	$\emptyset$	$\emptyset$
q	{q, r}	{r, s}	{r, s}	q	{q, r}	{p, q}	{p, q, r}
r	{r}	{r, s}	{r, s}	r	{r}	$\emptyset$	$\emptyset$
s	{r, s}	{r, s}	{r, s}	s	{r, s}	$\emptyset$	$\emptyset$
	ϵ^*	c			ϵ^*		
p	{p}	$\emptyset$			$\emptyset$		
q	{q, r}	{q, r}			{p, q, r}		
r	{r}	{q, r}			{p, q, r}		
s	{r, s}	{q, r, s}			{q, r, s}		

NFA

	a	b	c
→ p	{p, q, r}	∅	∅
*q	{r, s}	{p, q, r}	{p, q, r}
s	{r, s}	∅	{p, q, r}
r	{r, s}	∅	{p, q, r}

DFA

	a	b	c
→ p	q	t	t
*q	r	q	q
*r	r	q	q
t	t	t	t

